KUAN-YI LI (李冠毅)

Digital IC Design Engineer @ SCT Ltd. Taiwan Branch
TOEIC 890

Mobile: +886-952-765-169
E-mail: s51511830@gmail.com

Professional Experience

SCT Ltd. Taiwan Branch

Digital IC Design Engineer **2025.10 ~ Present**

Design and modify RTL modules using Verilog for digital IC systems.
Perform functional debugging and waveform analysis using simulation tools in Linux environment.
Support timing-aware RTL development and assist synthesis-related issue analysis.
Participate in internal bring-up and design validation flows.
Collaborate with cross-functional teams for feature integration and verification.

Selected Projects & Research Experience

Master's thesis:

Object-Location Prediction Based on CIE Color Difference for Deep Reinforcement Learning. **2023.10**

Proposed a lightweight method (OLP-CIE) for object location prediction using CIE color difference (CIEDE2000) to enhance DRL. Achieved 85.5% location accuracy and 90.4% object color identification accuracy. This work was presented at the 2023 GCCE held in Nara, Japan. (https://ieeexplore.ieee.org/document/10315341)

Algorithm and Architecture Design of Fast-Convergent Deep Reinforcement Learning for Object-Location Prediction **2025.01**

Designed a DRL algorithm (FCDQN) to improve object-location prediction efficiency, reducing training iterations by 76% while maintaining high localization accuracy.
Implemented the hardware architecture of a fast-convergent DRL accelerator in Verilog, verified functionality with ModelSim, and performed RTL-to-gates synthesis using Synopsys Design Compiler, APR and DRC using Synopsys IC Compiler, along with STA under TSMC 90nm technology, achieving ~75% energy savings and ~20% area reduction compared to prior DQN accelerators.

Education Background

Yuan Ze University M.S in Electrical Engineering (2025)
Yuan Ze University B.S in Electrical Engineering

Technical Skills

Domain expertise: Digital IC Design, RTL Development, Timing-aware Design.

Programming & Tools: Verilog, Synopsys Design Compiler, Synopsys IC Compiler, MATLAB, ModelSim, Linux.

Professional Training & Conferences

Cell-Based IC Physical Design and Verification with IC Compiler (21hr)
TSRI, 2023.03

2023 IEEE 12th Global Conference on Consumer Electronics (GCCE 2023)
Nara, Japan, 2023.10

2024 International Automatic Control Conference (CASA 2024)
Taoyuan, Taiwan, 2024.10

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CERTIFICATE OF ACHIEVEMENT

This is to certify that

LI KUAN-YI

achieved the following scores on the

TOEIC® Listening & Reading Test

Listening	485
Reading	405
Total	890

as administered under the auspices of: Chun Shin Limited
at: Taiwan
date: 17-August-2025

Updated TOEIC L&R
25509698

This is presented in recognition of your achievement.
An official score report must be submitted to organizations that require a TOEIC score.

TOEIC SCALE
Listening: 5-495
Reading: 5-495
Total: 10-990

Lucy Li
Executive Director
Global TOEIC Management, ETS

Letter of Recommendation

To Whom It May Concern,

I am pleased to write this letter of recommendation for **Mr. Kuan-Yi Li**, who completed his Master's degree in Electrical Engineering at Yuan Ze University under my supervision in 2025 July. During his graduate studies, I had the opportunity to closely observe his work, dedication, and problem-solving capabilities, particularly through his research on AI hardware acceleration and deep reinforcement learning.

Kuan-Yi is an independent thinker who is always willing to explore new areas. He actively engaged in research discussions, consistently found creative solutions to novel problems, and exhibited excellent communication and teamwork skills.

I am confident that his technical expertise, along with his strong work ethic and collaborative spirit, will make him a valuable asset to any team. I highly recommend Mr. Kuan-Yi Li for any position related to **Digital IC Design or AI algorithm development**.

Please feel free to contact me if further information is needed.

Sincerely,

Dr. Shu-Yen Lin
Associate Professor
Department of Electrical Engineering, Yuan Ze University
Email: sylin@saturn.yzu.edu.tw
Phone: 03-4638800 #7130

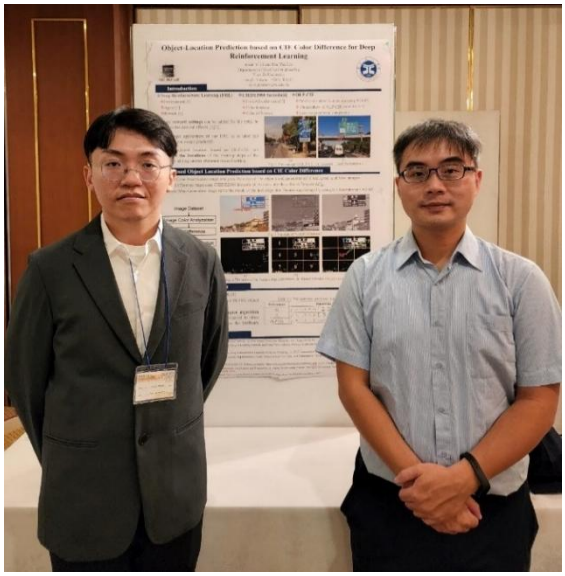
Yuan Ze University

By authority of the Board of Trustees of the University
and on the recommendation of the Faculty of
the Department of Electrical Engineering

has conferred on
LI, KUAN-YI 李冠毅
the degree of
Master of Science
With all of the rights, privileges and honors appertaining to that Degree
Given under the seal of the University in Taiwan
on the January in the year Two Thousand and Twenty-five

President | Dean

SN:1104622



IEEE Advancing Technology for Humanity
IEEE CTSoc CONSUMER TECHNOLOGY SOCIETY
GCCE 2023

Certificate of Presentation

to

Kuan-Yi Li

at the

2023 IEEE 12th Global Conference on Consumer Electronics
on October 10th – 13th, 2023
at Nara Royal Hotel, Nara, Japan

Ryota Nishimura
Conference Chair
IEEE GCCE 2023

NARLabs 國家實驗研究院

台灣半導體研究中心

Taiwan Semiconductor Research Institute

結業證書

(112) 課字-晶段-第 1121024 號

李冠毅 於中華民國 112 年 3 月 1 日至中華民國 112 年 3 月 3 日參加本中心「晶片設計課程-Cell-Based IC Physical Design and Verification with IC Compiler」共計 21 小時，特此証明。

主任 侯拓宏

中華民國 112 年 3 月 3 日

Taiwan Semiconductor Research Institute

Certificate of Completion

This is to certify that KUAN-YI LI has completed 21 hours Chip Design Course-Cell-Based IC Physical Design and Verification with IC Compiler (from 03/01/2023 to 03/03/2023).

Tuo-Hung Hou

Director General

03/03/2